

MATH4065 – Further Topics in R

Linear Models

Let us return to the Scottish hill race data in the MASS library. Type

```
library(MASS)
names(hills)<-c("distance","height.climbed","record")
attach(hills)
```

We can fit a regression line using

```
out<-lm(record~distance)
```

and the results, including the fitted line, tests for the existence of regression etc. are put into the R object called `out`. To get a summary of the results type

```
summary(out)
```

Is there evidence for the existence of regression here? There are many other useful things you can do with the output, eg. add the fitted regression line

```
plot(distance,record)
abline(out)
```

We can fit a better linear model by accounting for the height gain as well as the distance when predicting the record time.

```
out<-lm(record~distance+height.climbed)
```

Looking at the `summary` shows that the parameter estimates are -8.992 , 6.218 and 0.110 . We can extract the fitted values directly:

```
out$fitted.values
```

Let us now calculate the raw residuals and give a summary:

```
residuals=record-out$fitted.values
summary(residuals)
```

In the output from `summary(out)`, the residual standard error is the square root of the mean residual sum of squares

$$\sqrt{\frac{1}{d} \sum_{i=1}^n r_i^2}$$

where r_i is the i th residual and d , the residual degrees of freedom, is the sample size, n , minus the number of parameters (including the intercept term) which have been estimated. In this example, the residual degrees of freedom is $35 - 3 = 32$ and the residual standard error is 14.68 . Therefore the residual sum of squares, $\sum_{i=1}^n r_i^2$, can be calculated to be 6896.1 .

The output from `summary(out)` also gives R^2 and, on the last line, the F -statistic. This statistic tests for the existence of regression, i.e. it tests the null hypothesis that the distance and height.climbed coefficients are both zero, using the mean regression sum of squares divided by the mean residual sum of squares.

#Central limit theorem

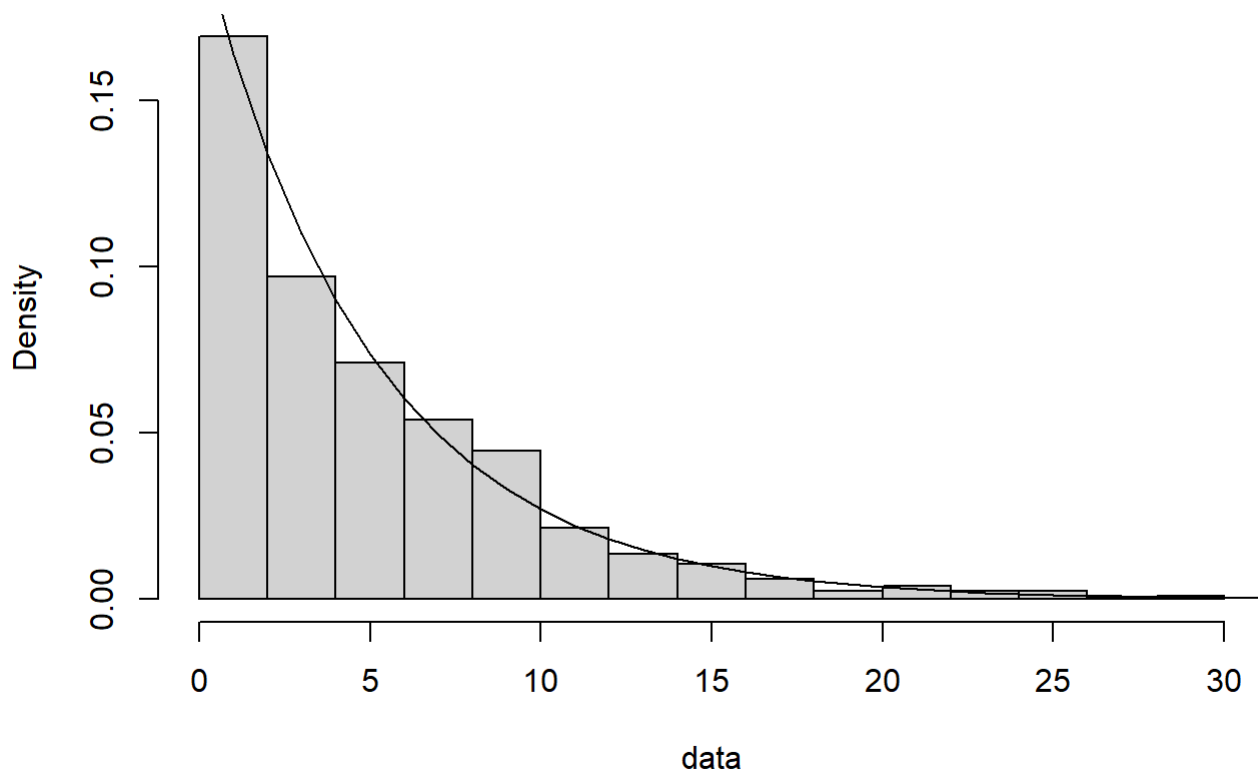
In this section we will illustrate the concept of the central limit theorem. Suppose that the service time for customers in a bank has an exponential distribution with mean 5 minutes. We can simulate 1000 such random service times using

```
data<-rexp(1000,rate=1/5)
```

We can plot a histogram of these times and overlay the pdf of the exponential distribution as follows

```
hist(data,freq=FALSE)
x<-0:40
y<-dexp(x,rate=1/5)
lines(x,y)
```

Histogram of data



The histogram and the pdf should have the same shape.

Exercise 1.

Calculate the sample mean, standard deviation, median and inter-quartile range (IQR) of the 1000 random service times.

Exercise 2.

Compare the sample mean, standard deviation, median and IQR with the corresponding theoretical values for this exponential distribution. Note that the `qexp` function may be helpful in finding the theoretical median and IQR.

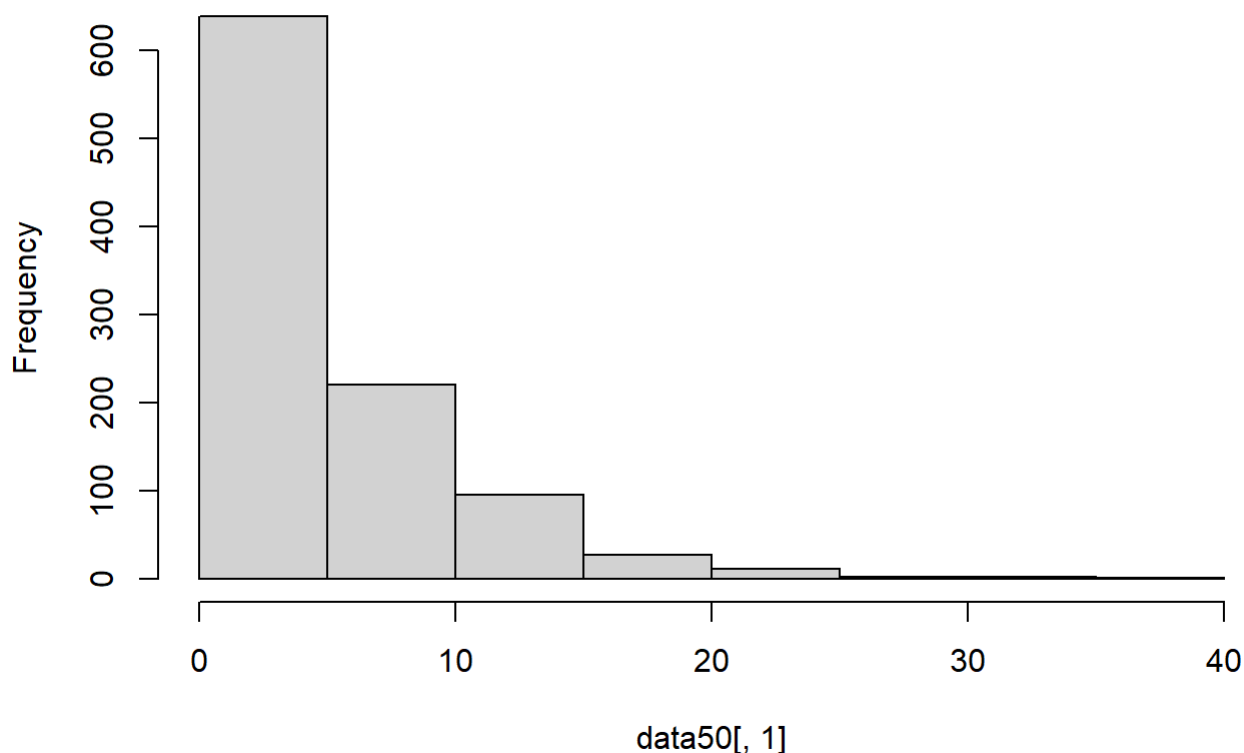
We will now consider the distribution of the mean service time per customer for 50 successive customers. Type the following

```
data50<-matrix(rexp(1000*50,rate=1/5),nrow=1000,ncol=50)
```

This generates 50000 numbers from the exponential distribution and places them in a large matrix with 1000 rows and 50 columns. We can think of each row being a different bank and each column representing the first, second, ..., fiftieth successive customer in that bank. Produce a histogram of the data in the first column using

```
hist(data50[,1])
```

Histogram of data50[, 1]



This should look like our earlier histogram. Try plotting histograms of the other columns to convince yourself that they have an exponential distribution. We can now calculate the mean service time per customer at each bank using

```
meantime<-rowMeans(data50)
```

Exercise 3.

Examine the distribution of these means by plotting a histogram and calculating summary statistics as before. Are there any major changes from before? The histogram should have the shape of a normal distribution. Using your knowledge of the central limit theorem, edit the following commands to overlay the pdf of a normal

distribution with an appropriate mean and standard deviation.

```
hist(meantime,freq=FALSE)
x<-seq(2,8,0.1)
y<-dnorm(x,mean=??,sd=??)
lines(x,y)
```

Exercise 4.

Suppose the random values were generated from a Cauchy distribution (`rcauchy`) instead of the exponential distribution. What happens to the distribution of the sample mean as the sample size increases?

Confidence intervals

In this section we will illustrate the concept of confidence intervals. We will generate 100 datasets, each with $n = 20$ observations from a $N(\mu, \sigma^2)$ distribution where $\mu = 0$ and $\sigma^2 = 100$.

```
mu<-0
sigma<-10
n<-20
data<-matrix(rnorm(1000*n,mean=mu,sd=sigma),nrow=1000,ncol=n)
```

For each dataset (row), we will calculate the sample mean.

```
m<-rowMeans(data)
```

Exercise 5.

If $X_1, \dots, X_n \sim N(0, 100)$, then what is the theoretical distribution of $\bar{X}$? Do the values in `m` agree with this? Edit the following commands to check your answer.

```
hist(m,freq=FALSE)
x<--10:10
y<-dnorm(x,mean=??,sd=??)
lines(x,y)
```

We will now calculate a 95% confidence interval for μ for each dataset. We will calculate the lower limit, `L`, and upper limit, `U`, for each interval as follows

```
alpha<-0.05
L<-m-qnorm(alpha/2,lower.tail=FALSE)*sigma/sqrt(n)
U<-m+qnorm(alpha/2,lower.tail=FALSE)*sigma/sqrt(n)
```

Note that `qnorm(0.025,lower.tail=FALSE)` is the table value $z_{0.025}$. Make sure you are clear how the above commands correspond with the formula for a confidence interval given in the notes.

Exercise 6.

How many of the 1000 95% confidence intervals would you expect to include the true value of μ ? Is this what you observe? You can use the following command to see if μ lies between the lower and upper limit for each sample.

```
L<mu & mu<U
```

Suppose we don't know the true value of σ . We can calculate the sample standard deviation for each dataset (row) as follows.

```
s<-apply(data,1,sd)
```

Exercise 7.

Adapt the commands given above to calculate 95% confidence intervals for μ assuming we don't know σ . Your interval should be based on the t -distribution. How many of the 95% confidence intervals would you expect to include the true value of μ ? Is this what you observe?

Exercise 8.

The commands above use the values $\mu = 0$, $\sigma^2 = 100$, $n = 20$ and $1 - \alpha = 0.95$. Try changing these values and repeating the above exercises with different values. (If you have written the commands in a script file then it should be relatively quick to do this by simply editing a few lines of the script file.)

In reality, of course, we usually only observe 1 dataset rather than 1000, so we can never be sure whether the confidence interval for the observed data includes the true value of μ . However, hopefully this exercise has helped to reinforce the concept that it is the interval which is random and not μ .

What next?

R is an extremely useful statistical tool and is used by students and professional statisticians alike. It is open-source software and new functions are being added to it daily as new statistical methods are developed. In these computer classes we have just scraped the surface of what R is capable of. Other statistics modules will use R too so it is worth investing time learning to use it. There are also many online tutorials if you want to extend your skills further.